

2nd Sem.1st Year.

Department of English  
End Semester Examination  
2022

(For the 2nd Semester Students of all the Departments of the Engineering Faculty)

ENGUGHU01: Communicative English

Full Marks: 80

Time: 3 Hours

Section I

1. Answer ANY TEN of the following questions:

2x10=20 Marks

- A. What is code?
- ☒ B. What is extra-personal communication?
- ☒ C. Exemplify a Grapevine communication.
- ☒ D. What is feedback in communication?
- ☒ E. What is *lingua franca*?
- F. Give an example of paralinguistic sound used in communication.
- ☒ G. Give an example of Horizontal communication.
- ☒ H. What is the basic difference between general communication and professional communication?
- I. What is semantic gap?
- J. What is passive listening?
- ☒ K. Exemplify a barrier in effective listening.
- L. How many diphthongs are there in English language?
- ☒ M. What is phoneme?
- ☒ N. What is skimming in reading?
- O. Give two examples of idioms in English language.
- ☒ P. What is modal verb? Give an example.

**Section II****5x4=20 Marks**2. Answer **ANY FOUR** of the following questions: (word limit 200)

- A. Distinguish between formal and informal communication.
- B. Define body language. Mention some of the gestures we use to communicate.
- ☒ C. What are the benefits of effective listening?
- ☒ D. What is monologue?
- ☒ E. Write a short dialogue between two friends discussing the price hike.
- F. What are the important features of close reading?
- ☒ G. What is a report? Write the characteristics of a good report.

**Section III****10X4=40 Marks**3. Answer **ANY FOUR** of the following questions. (word limit 400)

- ☒ A. Define communication. What are the functions of communication? (5+5)
- B. Identify the notable barriers of an Effective Communication. Discuss the means to overcome those barriers. (6+4)
- C. Define Organizational communication. What are the demerits of Grapevine communication? (5+5)
- ☒ D. Attaching a resume, write a job application for the post of a teacher in a school. (10)
- ☒ E. Define and discuss the differences between verbal communication and non-verbal communication. (10)
- F. Mention the different presentational strategies of Speaking Skills. (10)
- ☒ G. You are the sports secretary of your school. Write a letter to Globe Enterprises, a leading firm dealing in sports goods, requesting them to supply their trade catalogue. You may mention the items you intend to buy and ask for a discount on the catalogue prices. (10)



ALIAH UNIVERSITY  
DEPARTMENT OF MATHEMATICS AND STATISTICS  
SPRING SEMESTER END EXAMINATION, 2022  
PAPER MATUGBS02 ( ENGINEERING MATHEMATICS - II)

Full Marks 80

Time 3 hrs.

Use separate answer scripts for each group.

Group – A (40 Marks)

Answer any eight questions:- (5x8=40)

1. Express  $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{pmatrix}$  as the sum of a symmetric and a skew symmetric matrix.

2. Show that  $\begin{vmatrix} 1 & a & a^2 \\ a^2 & 1 & a \\ a & a^2 & 1 \end{vmatrix}$  is a perfect square. Verify the result for  $a = 3$ .

3. Find characteristic equation and eigenvalues of the matrix  $\begin{pmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{pmatrix}$ .

4. State Cayley Hamilton theorem. Use Cayley Hamilton theorem to find  $A^{50}$ , where  $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ .

5. Show that  $\begin{vmatrix} 1 & \cos x - \sin x & \cos x + \sin x \\ 1 & \cos y - \sin y & \cos y + \sin y \\ 1 & \cos z - \sin z & \cos z + \sin z \end{vmatrix} = 2 \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix}$ .

6. If  $a, b, c$  are real numbers, prove that  $a^2 + b^2 + c^2 \geq ab + bc + ca$ .

7. Apply elementary row operation to reduce the following matrix to a row echelon matrix

$$\begin{pmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{pmatrix}$$

8. Solve the following system of equations by Cramer's method

$$x + 2y - 3z = 1$$

$$2x - y + z = 4$$

$$x + 3y = 5$$

$$2x - y + z = 4$$

9. Find the inverse of  $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 4 & 4 \\ 3 & 3 & 7 \end{pmatrix}$ .

10. Apply matrix method to show that  $x = 3, y = 1, z = 2$  is a solution of the system of equations  $2x + 3y + z = 11, x + y + z = 6$  and  $5x - y + 10z = 34$ .

11. If  $\alpha, \beta, \gamma$  are roots of the equation  $x^3 + px^2 + qx + r = 0$ , find the value of  $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$  in terms of  $p, q$  and  $r$ .

12. Prove that the intersection of two subspaces of a vector space  $V$  over a field  $F$  is a subspace of  $V$ .

13. Find  $\text{mod } z$  and  $\text{arg } z$  and express  $z$  in polar form where  $z = -1+i$ . (1+1+3)

14. Prove that the set of vectors  $\{(1,2,2), (2,1,2), (2,2,1)\}$  is linearly independent in  $\mathbb{R}^3$ .

15. Find the rank of  $\begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 8 & 6 \\ 3 & 6 & 6 & 3 \end{pmatrix}$ .

16. When is an  $n \times n$  matrix called a row reduced Echelon Matrix? Give example of it.

#### Group – B(40 Marks)

1. Answer the following questions: 1X5 = 5

(a) Under what condition the following differential equation  $(2ax + by + g)dx + (2fy + cx + h)dy = 0$  is exact?

(b) Find the integrating factor of the differential equation :

$$x \cos x \frac{dy}{dx} + (x \sin x + \cos x)y = 1.$$

(c) Find the complementary function of the differential equation:

$$\frac{d^3y}{dx^3} + 2 \frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{2x}.$$

(d) Find the complementary function of the differential equation:

$$\frac{d^2y}{dx^2} - 4y = \sin 2x.$$

(e) If  $y_1 = e^x$  and  $y_2 = e^{2x}$ , then find the Wronskian  $W(y_1, y_2)$  of  $y_1$  and  $y_2$ .

2. Solve ANY THREE of the following differential equations: 5X3=15



(a)  $(y^2 e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0.$

(b)  $3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0.$

(c)  $(y^4 - 2x^3 y)dx + (x^4 - 2xy^3)dy = 0.$

(d)  $\frac{dy}{dx} = \frac{y-x+1}{y+x+5}.$

(e)  $x(1-x^2)\frac{dy}{dx} + (2x^2-1)y = ax^3$ ,  $a$  being a fixed constant.

(f)  $\frac{dy}{dx} + \frac{y}{x} \log_e y = \frac{y}{x^2} (\log_e y)^2.$

3. Solve **ANY TWO** of the following differential equations:

5X2=10

(a)  $\frac{d^3 y}{dx^3} - 3\frac{d^2 y}{dx^2} + 3\frac{dy}{dx} - y = xe^x + e^x.$

(b)  $\frac{d^2 y}{dx^2} - 2\frac{dy}{dx} + 2y = 5 \cos x$ , given that  $y = \frac{dy}{dx} = 0$ , when  $x = 0$ .

(c)  $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - y = 3e^x \cos(\log_e x).$

(d)  $x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = x^4.$

4. Answer **ANY ONE** of the following questions:

6+4=10

(a) (i) Using the Method of variation of parameters, find the general solution of the differential equation:  $\frac{d^2 y}{dx^2} + a^2 y = \tan ax$ .

(ii) Find the eigenvalues  $\lambda_n$  and the corresponding eigenfunctions  $y_n$  for the differential equation  $\frac{d^2 y}{dx^2} + \lambda y = 0$  ( $\lambda > 0$ ) satisfying the boundary conditions,  $y(0) = 0$  and  $y(\pi) = 0$ .

(b) (i) Using the Method of Undetermined Coefficients, find the general solution for the differential equation:  $\frac{d^2 y}{dx^2} - 4\frac{dy}{dx} + y = x^2 - 2x + 2$ .

(ii) Find the eigenvalues  $\lambda_n$  and the corresponding eigenfunction for the differential equation  $\frac{d^2 y}{dx^2} + \lambda y = 0$  ( $\lambda > 0$ ) satisfying the boundary conditions  $y(0) = 0$  and  $\frac{dy}{dx} = 0$  at  $x = \pi$ .



## Answer any five questions

5x16=80 Marks

1. a) Define i) charge ii) electric current iii) power iv) network 4 Marks  
 b) State the steps to solve an electric circuit problem by the Thevenin's Theorem. 3 Marks  
 c) For the circuit shown in Fig. 1 find  $V_{CE}$  and  $V_{AG}$  4 Marks



Fig. 1

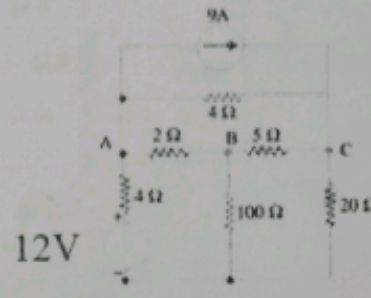


Fig. 2

- d) Use nodal analysis to determine the voltage across 5 ohm resistance and the current in the 12V source. (Refer. to Fig. 2) 5 Marks
2. a) What is an ideal transformer? Derive the condition of maximum efficiency for a single phase transformer using common notations. 5 Marks  
 b) In a 50 kVA 2000/200V, 50 Hz single phase transformer, the iron and copper losses are found to be 800 W and 400 W at the full load and unity p.f, respectively. Calculate the efficiency of the transformer with same power factor at i) full load and ii) one third of load. 6 Marks  
 c) Draw the phasor diagrams for different type of loads in an ideal transformer. 5 Marks
3. a) Define phasor and phase angle. Explain different phasor for a single phase circuit with relevant diagrams. 6 Marks  
 b) An alternating voltage is given by  $V=230\sin 314t$ . Calculate i) frequency, ii) maximum value, iii) average value, iv) RMS value. 4 Marks  
 c) A series RLC circuit has  $25\Omega$  resistance, 0.2 mH inductance and  $66.3\mu\text{F}$  capacitance and is connected to a 240V, 60Hz supply. Find (i) the current, (ii) the power factor and (iii) the voltage drop across each element of the circuit. 6 Marks
4. a) Draw the connection diagrams of D.C Generator for different type of excitations. 5 Marks  
 b) A DC long shunt compound generator delivers a load current of 10A at 420V. The resistance of the armature, series field and shunt field are  $0.03\Omega$ ,  $0.015\Omega$  and  $150\Omega$ , respectively. Calculate the emf induced in the armature. Allow a brush voltage drop of 1V/brush. 6 Marks  
 c) Sketch the O.C.C of D.C shunt generator. 2 Marks  
 d) What is the significance of back EMF in a DC motor? 3 Marks
5. a) Derive the condition for maximum power transfer from a source to resistive load. 5 Marks  
 b) What is DC motor? Derive the expression of torque for a DC motor. 6 Marks  
 c) Write a short note on series resonance. 5 Marks

6. a) State Norton's theorem and find current using Norton's theorem through a load of 8 ohm in the circuit shown in Fig. 3
- b) In the network shown in Fig. 4, find the value of R such that maximum possible power will be transferred to  $R_L$ . Find also the value of the maximum power and the power supplied by source under this conditions. 2x8=16 Marks

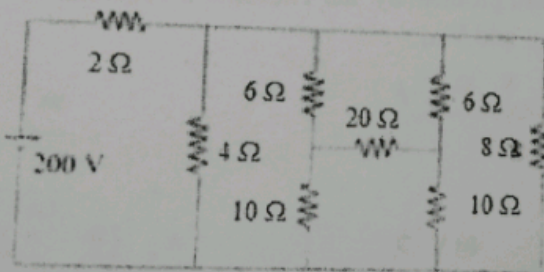


Fig. 3

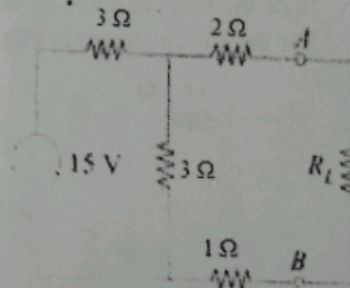


Fig. 4



# Aliah University

End-Semester Examination (Spring Semester) - 2022

B.Tech. (CEN/MEN) 1st Year 2nd Semester

Paper Name: Programming For Problem Solving

Full Marks: 80

Paper Code: CSEUGES01

Time: 3 hrs

## Group-A

(Answer any four questions)

4 × 5 = 20

1. What is the purpose of printf() and scanf() in C Program?
- ✓ 2. Write a short note on flowchart.
3. Explain the ternary operators with examples.
4. State the differences between a do-while loop and a while loop.
- ✓ 5. What is an array?
- ✓ 6. Discuss about pointer expressions.

## Group-B

(Answer any six questions)

6 × 10 = 60

1. (a) Write a flowchart to compute the area of rectangle. 5  
(b) Write an algorithm to compute the factorial of a given number. 5
2. ✓ (a) Discuss different types of logical operators. 5  
(b) Write a C program to show the use of relational operators. 5
3. (a) ✓ Discuss different types of iterative statements. 5  
(b) Write a C program to calculate the sum and average of first n numbers. 5
4. (a) ✓ Explain the importance of switch case statements. 4  
(b) Write a C program to create a simple calculator using a switch case statement. 6
5. ✓ (a) How are functions defined? 4  
(b) Write a C program to find the smallest of three numbers using a function. 6
6. (a) What do you mean by call by value and call by reference? 4  
(b) ✓ Write a C program to swap the contents of two variables. 6
7. (a) ✓ How two-dimensional arrays are declared. 3  
(b) Write a C program to multiply two matrices. 7
8. (a) ✓ How are strings read? 4  
(b) ✓ Write a C program to find the length of a string without using a library function. 6



## ALIAH UNIVERSITY

Name of the Examination: Semester(II)

Session: 2021-22

Subject: Engineering Chemistry

Course code: CHMUGBS01

Duration: 3 Hrs

Full Marks: 80

(Use Separate Answer Sheet for Each Group)

## Group A

1. Explain the following terms (*any TEN*): (i) State function (ii) Adiabatic process (iii) Molar heat capacity (iv) Reversible expansion (v) Enthalpy (vi) Intensive property (vii) Path function (viii) Internal energy (ix) Isochoric process (x) Homogeneous system (xi) Isolated system (xii) Cyclic process 2.5x10

Answer any Three(2)

2. (a) Deduce the equation of work for reversible isothermal expansion of an ideal gas. (b) Prove that work done is more in reversible process than that of irreversible process. 2.5+2.5
3. (a) Establish the relation between  $\Delta H$  and  $\Delta U$ . (b) For the reaction  $H_2F_2(g) \rightarrow H_2(g) + F_2(g)$ . Calculate  $\Delta H$  for the reaction. 2.5+2.5
4. (a) Derive  $C_p \cdot dq_p/dt$  and  $C_v \cdot dq_v/dt$ . (b) Establish their relation for the ideal gas. 2.5+2.5
5. (a) What is Joule Thomson phenomenon? (b) Prove that for the Joule Thomson phenomenon,  $\Delta H < 0$ . 2.5+2.5

## Group B

9x5= 45

Answer any Nine (9)

6. Write short notes on:

- (a) Bakelite  
(b) Teflon

7. Give a brief account of:

- (a) Vulcanization of rubber  
(b) Polythene

8. Explain the following with suitable reactions

- (a) Phenol formaldehyde resin  
(b) Polystyrene

9. How are the following polymers prepared? Mention their uses

- (a) PVC  
(b) LDPE

10. Write informative notes on the mechanism of radical polymerisation with suitable example

11. (a) What is meant by degree of polymerization?

- ✓ (b) Distinguish between addition & condensation polymerisation with example
12. (a) How is the crepe rubber obtained from latex ?  
(b) Give the structural unit of gutta percha
- ✓ 13. Why are silicones called inorganic polymers? Discuss the synthetic procedure of linear chain silicones
14. Write note on conducting polymers with reference to oxidative or p-doping with mechanism
15. (a) How will you prepare Styrene-Butadiene Rubber (SBR). Mention its uses  
(b) What is Ziegler-Natta catalyst ?
16. Write short notes on:  
✓ (a) Compounding of rubber  
(b) Conducting polyaniline
- ✓ 17. What do you mean by green chemistry. Give its importance
18. (a) What is meant by degree of polymerisation ?  
(b) How will you prepare polystyrene using anionic addition polymerisation method ?